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Trajectory planning for unmanned aerial vehicles in complicated urban environments: A control network approach☆

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ABSTRACT

Trajectory planning is crucial for Unmanned Aerial Vehicles (UAVs) applications. It is still a challenge to plan UAV trajectories in large-scale complicated three-dimensional urban environments, mainly due to the complexities associated with handling a tremendous number of obstacles. To relieve this difficulty, this study proposes a two-stage control network approach. The first stage is an offline planning stage, aiming at constructing a discrete control network in the working environment achieving collision avoidance and meanwhile considering UAV kinetic properties. The second stage is an online planning stage, where a feasible corridor is generated based on the shortest path in the control network, and the obtained route is smoothed by a three-dimensional Dubins curve algorithm; then, a time-optimal trajectory generation algorithm is applied to trace the designed spatial path. Numerical studies based on a virtual city as well as a large-scale real-world urban area are conducted. The results show that the proposed approach is able to identify reasonable UAV trajectories within negligible computational time.

1. Introduction

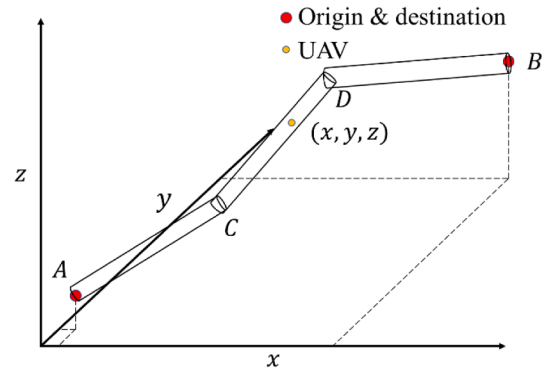
In recent years, unmanned aerial vehicles (UAVs) have attracted increasing attention in time-demanding urban mobility applications. Among those creating the most promising applications, many major e-commerce companies, including Amazon, DHL, are developing their own UAV delivery systems (DHL, 2014; Amazon, 2016). Through local distribution services, multiple drones can be used for extensive customer deliveries in the service area, and the total reduction in delivery time by using UAVs provides users with faster delivery services (Perboli and Rosano, 2019; Salama and Srinivas, 2020). Additionally, compared to conventional delivery modes, the implementation of UAV systems can save labor costs and reduce unreliability due to surface transportation. On the other hand, it has been proposed that UAVs can be used for fast-response applications to emergent urban incidents (Macias et al., 2020). For example, when a traffic accident is detected, UAVs nearby can be deployed quickly to respond to the information and rush to the scene to provide first-hand data for incident management, without causing surface traffic congestion. Other potential applications of UAVs can be found in health emergency services (Michael and Francisco, 2016), mobile data-collection (Barmpounakis and Geroliminis, 2020; Chen et al., 2020; Wan et al., 2020), communication networks (Macrina et al., 2020), and so on.

It can be noted from these potential UAV applications that service efficiency is a key focus. To improve efficiency, there are two main directions: the first is to further enhance the mechanical properties of the UAVs, and the second is to appropriately plan the flying

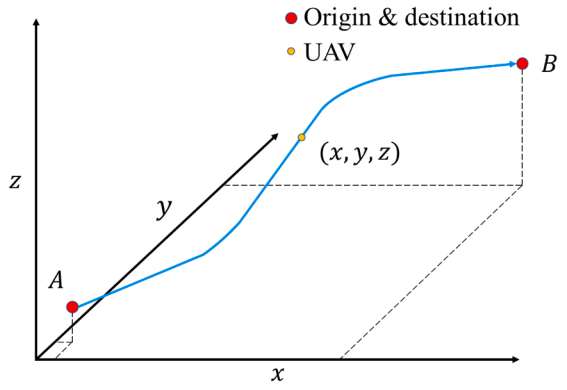
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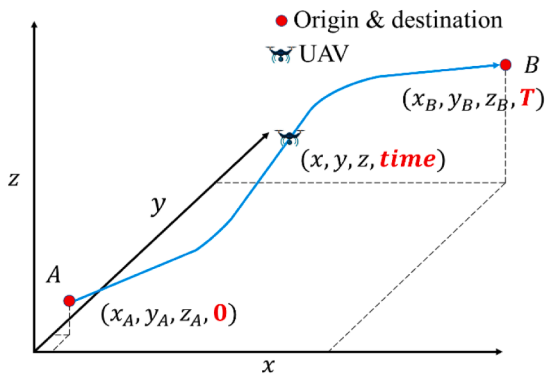
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(a) Illustration of a route



(b) Illustration of a path



(c) Illustration of a trajectory

Fig. 1. Illustration of three different terminologies used.

trajectories of the UAVs fulfilling given tasks. This study mainly focuses on the latter direction. Trajectory planning plays an important role in the majority of UAV studies. For a given task with an origin and a destination, from the spatial perspective, the trajectory connects the start point and end point and avoids obstacles, generating a feasible 3D geometric path. In addition, given a spatial path, from a temporal perspective, we need to decide the detailed motion plan that allows the UAV to fly the path with minimum travel time. In contrast to the traditional 2D trajectory planning problem for autonomous vehicles, 3D trajectory planning for UAVs has considerably more complexities for two main reasons. First, the feasible space for the operation of UAVs is a continuous 3D space containing infinitely feasible trajectories with a wide variety of shapes; second, in some common working environments (especially in urban areas), obstacles (buildings, trees or lamp posts) further complicate the structure of the feasible space, making the trajectory planning

problem more difficult to solve.

Existing studies on 3D UAV trajectory planning problems can be generally classified into 3 categories: sampling-based methods, heuristic-based methods and two-stage methods. We provide a brief review of the related literatures in the following.

The best-known sampling-based methods include Rapid-exploration Random Tree (RRT) algorithm along with its improved version, i.e., RRT*, and Probability Road Map (PRM). The RRT algorithm (LaValle, 1998, Adiyatov and Varol, 2013) expands a tree structure from the starting point, and the expansion direction of the tree is determined by randomly sampling points in the probing space. However, the path obtained by RRT is a random feasible path rather than an optimal path. RRT* algorithm (Li et al., 2017, Nurimbetov et al., 2017) adjusts the tree structure while expanding nodes to maintain optimal structure, and iteratively finds the optimal path. In order to confine the expanding space, Informed RRT* (Gammell et al., 2014) was proposed by only sampling in an ellipse space with kinodynamic characteristics. On the other hand, based on graph search, PRM (Kavraki et al., 1996; Aggarwal and Kumar, 2020) performs probabilistically complete path planning based on simplified graphs, which is relatively efficient for collision checking. Sampling-based approach can only find a spatial path, and executing the flight process requires further smoothening the path (e.g., by B-spline method, Shi et al. 2020) and determining the motion of UAVs along the path.

The second category is heuristic-based algorithms; they employ customized heuristics to acquire feasible paths from origin to destination, and with sufficient computational time the flying time associated with the best feasible path explored is expected to be near optimal. The most influential heuristic-based methods include Ant Colony Optimization algorithm (ACO) (Ma et al., 2007), Genetic Algorithm (GA) (Gao et al., 2005), and Particle Swarm Optimization algorithm (Huang and Fei, 2018). Dynamic models and kinematic constraints including flying speed, climb angle rate are taking into consideration on the basis of GA (Gao et al., 2005, Bagherian and Alos, 2015, Azadeh et al., 2019).

As the third line of research, i.e., the two-stage methods, has received more attention as the complexity of path planning problem evolves (Hwangbo et al., 2007; Azadeh et al., 2019). The two-stage methods decompose the overall trajectory determination problem into two stages; the first stage identifies the collision-free kinodynamic compatible path in the 3-dimensional space, and then in the second stage it solves the optimized trajectory corresponding to the discretization time in the flight path. For the first stage problem, it still faces the difficulty of finding a sufficiently good path in a complicated working environment, so some simplification mechanisms are still required.

Other than the above methods, there are also studies focusing on the extensions of trajectory determination problem by incorporating the randomness or dynamic obstacles in the flight process; representative methods in handling such problems include Markov Decision Process (MDP) approach, Model Predictive Control (MPC) approach, and Potential Field Method (PFM). MDP (Puterman, 1994; Darbari et al., 2017) models the stochasticity via discrete-state Markov decision process, such that the action is dynamically determined according to the current state and a probability matrix of state transition. For MPC approach, the stochastic environment can be modeled in a more sophisticated manner, and the real-time decision is made via solving a rolling-horizon model, often formulated as a mathematical program (Yao and Zhao, 2015). On the other hand, the PFM simulates the environment with a potential field (e.g., a thermal field), and utilizes some simple algorithm (e.g., fast-marching algorithm, Garrido et al., 2006), to find a path with minimal resistance within the field. This type of method can handle cases with dynamic obstacles (Ge and Cui, 2002).

With respect to complicated urban environments, the trajectory planning problem for UAVs is extremely difficult. In a metropolitan area, there are generally a large number of buildings of various sizes and heights, and these buildings constitute an intricate set of obstacles; in addition, many UAV tasks require travel over long distances (and sometimes the heights of the origin and the destination differ), suggesting that feasible trajectories are often associated with nonregular shapes. In these settings, both sampling-based and heuristic-based methods show serious limitations. With sampling-based and heuristic-based methods, they may be difficult to identify a feasible trajectory from the origin to destination in a complicated environment, and the computational time often becomes unacceptable. Additionally, even with sufficiently long computation time, the solution qualities of both types of methods are difficult to guarantee. Considering kinetic properties of UAVs in the search algorithm is also of great importance for identifying a reasonable trajectory.

To fill the research gap, this paper proposes a novel methodological framework to identify a reasonable trajectory solution within an acceptable computational time. The trajectory planning framework has two stages, a network establishment stage and an online trajectory planning stage. In the offline stage, we construct a discrete control network (along with its expansion) as the basis for UAV trajectory planning. The control network is composed of a set of nodes and directed links, with the link costs evaluated by a kinetics-related calculation of the UAVs. In this paper, we define three terminologies to represent the results corresponding to different stages of the algorithm: i.e., *route*, *path*, and *trajectory*. As shown in Fig. 1(a), a *route* in the control network represents a valid "corridor" for a feasible trajectory, and no obstacle is presented along the corridor. Based on the control network, the online stage generates an optimal piecewise linear curve for a UAV given the origin and destination using a simple shortest-path algorithm, the "corridor" is obtained by replacing all linear segments on the curve with cylinders of identical length. With the corridor suggested by the acquired route, we implemented the Dubins curve smoothing technique (Dubins, 1957; Shanmugavel et al., 2013; Stefano et al., 2018) to obtain a sufficiently smooth spatial *path* for the UAV that ensures collision avoidance (Fig. 1(b)). Finally, a time-optimal trajectory generation algorithm is applied to trace the designed spatial path, and acquires a *trajectory* for the UAV; in other word, a trajectory is a path with time dimension to capture the UAV spacetime movement (Fig. 1(c)). Compared with sampling-based and heuristic methods for the 3D UAV trajectory planning problem, our control network-based strategy is able to provide near-optimal solutions with significantly less computational effort in a complicated urban environment. In essence, we decomposed a large-scale infinite-dimensional nonlinear optimization problem into a discretized network routing problem, a curve smoothing problem, and a time-optimal trajectory design problem given a fixed path, and the latter three problems were all substantially easier than the original one. Numerical studies based on artificial and real-world cases demonstrated the effectiveness of the proposed methodological framework. The established control

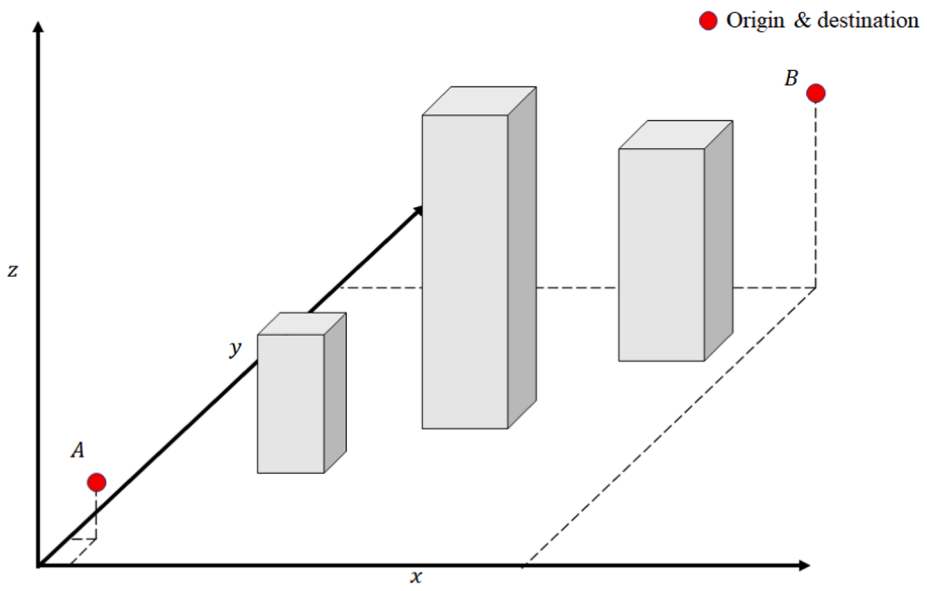


Fig. 2. An urban area showing the UAV origin and destination.

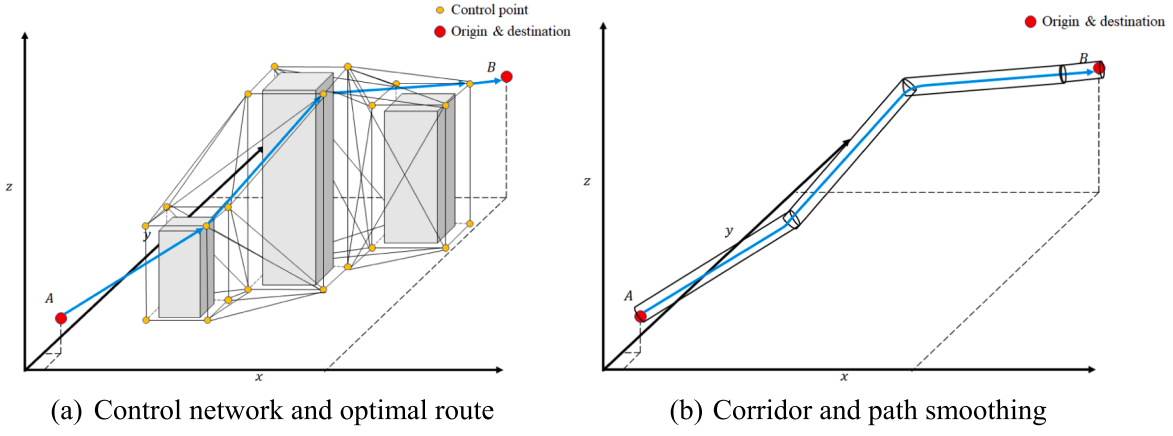


Fig. 3. Control network and path smoothing.

network can also be treated as the basis to support further time-sensitive UAV applications in future cities, such as for parcel delivery or service patrols.

The remainder of the paper is organized as follows. Section 2 introduces the methodological framework, including control network establishment, path curve smoothing and time-optimal trajectory generation. Section 3 conducts numerical tests on the proposed methodology with one virtual urban area and one real-world urban area. Finally, Section 4 concludes the whole paper. Three terminologies used in this paper are defined: *route* refers to a set of consecutive links in the control network; *path* refers a spatial curve plan that the UAVs follow; and *trajectory* is a spatiotemporal concept to represent the time–space curve planned for UAVs.

2. Methodology

2.1. Framework

In this subsection, we introduce the overall framework of our proposed methodology. The conceptual urban area considered is shown in Fig. 2, and our goal is to find the time-optimal trajectory from origin A to destination B that minimizes the total flying time of the UAV when tall buildings are in the path. The set of trajectory choices is infinite, and to alleviate this issue, we propose a two-stage methodological framework, with the first stage being offline and the second stage being online.

In the first (offline) stage, we construct a virtual *control network* for the city (see Fig. 3(a)). The control network is composed of a set of control points (or control nodes, labeled with yellow circles) and a set of links connecting these control points. We call a *route* in the

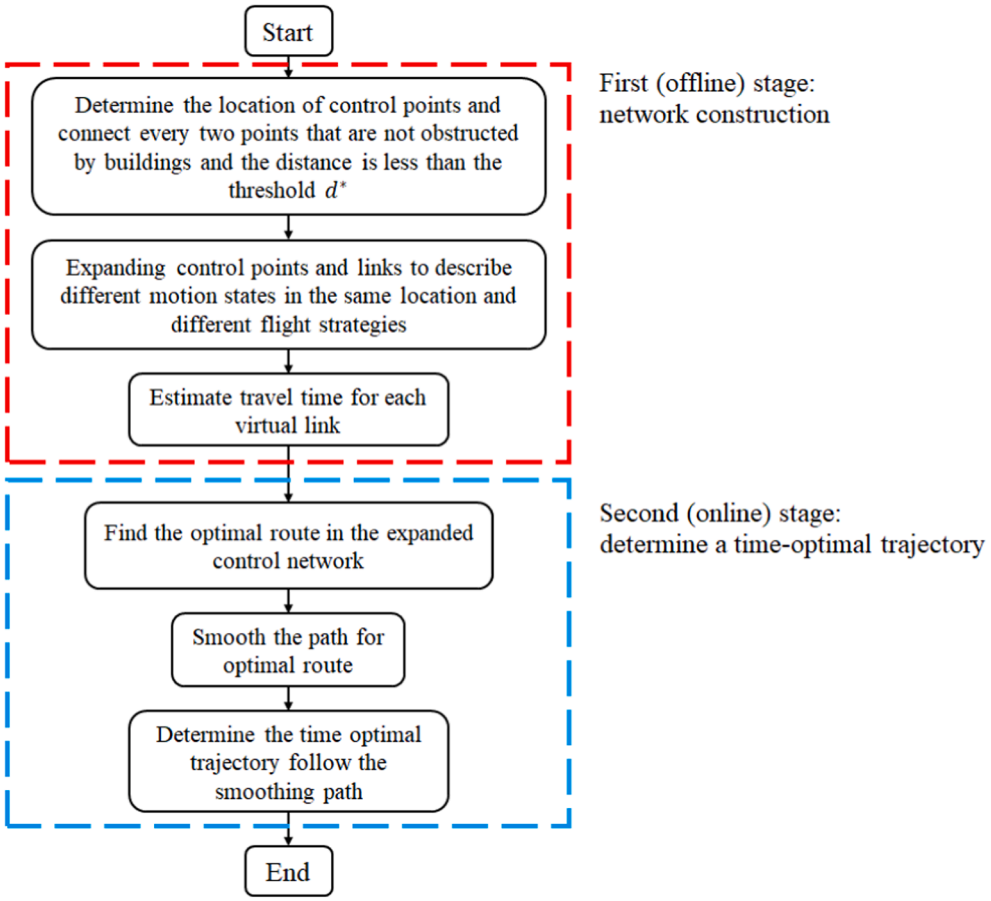


Fig. 4. Flow chart of the methodological framework.

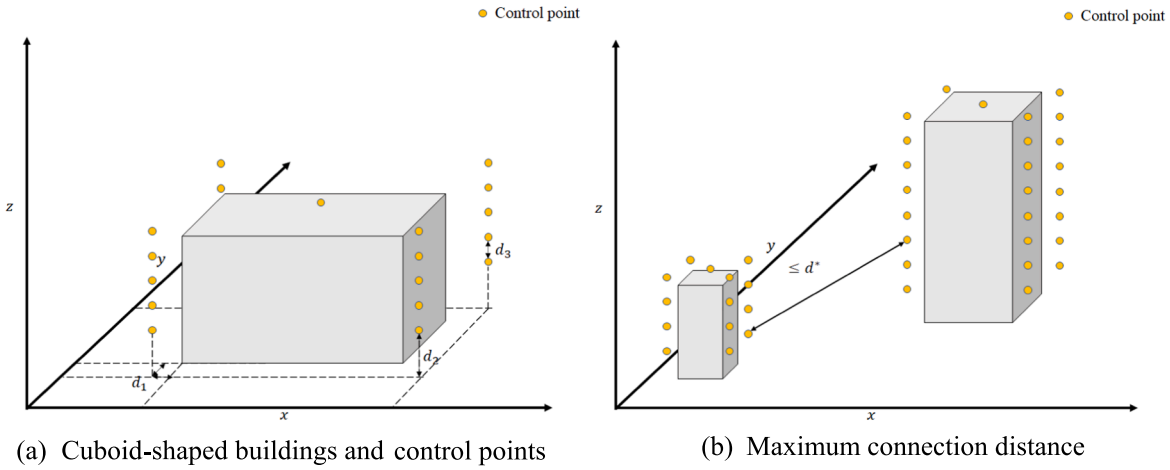


Fig. 5. Control points and link connection.

control network a set of consecutive links connecting the origin and the destination. All UAV movements are restricted to ensure that the UA is near a certain route in the control network; therefore, the number of possible paths becomes finite. With the control network, we can estimate the travel time for each link, with which we can adopt shortest-path algorithms to determine an optimal route. Obviously, the position and density choices of the control points and link connection establishment will affect the solution quality, and we analyze these impacts in the numerical experiment section.

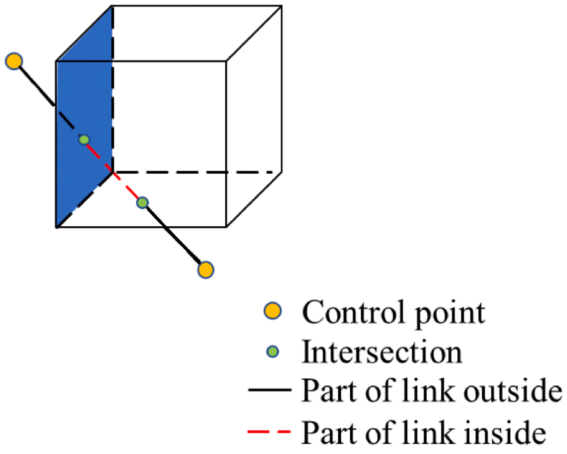


Fig. 6. Judging the collision between a link and a building.

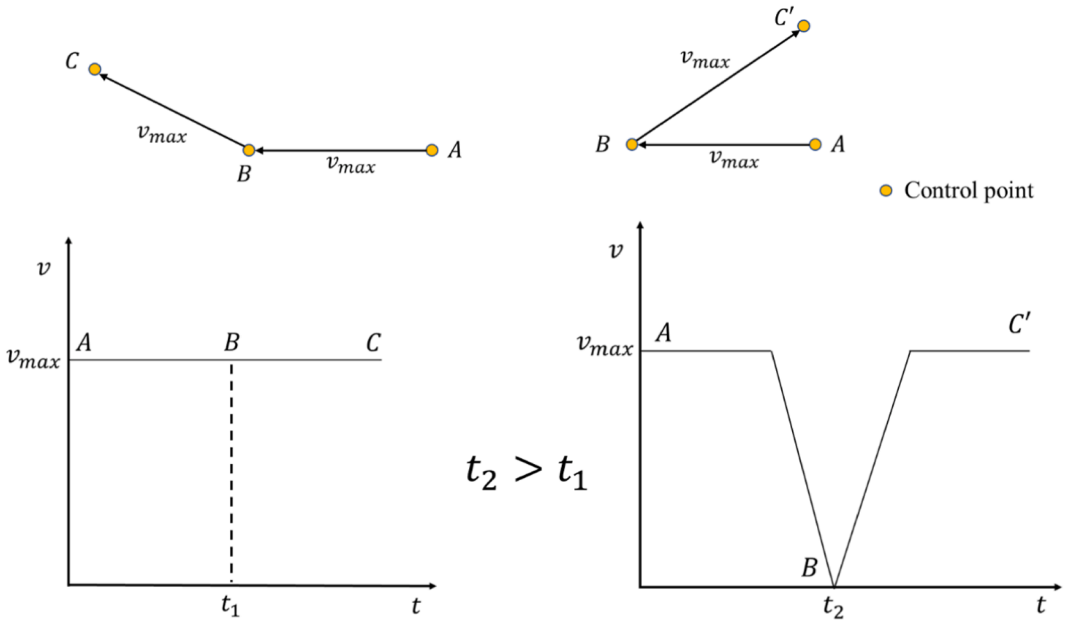


Fig. 7. Different flying strategies on the same link with different subsequent links.

After establishing the control network, we generate the second (online) stage in which the system determines a time-optimal trajectory based on the control network for one UAV given its origin and destination. The optimal route in the control network is computed for the UAV first (the blue line in Fig. 3(a)). Then, we treat the route as a "corridor" with a designated width, and within the corridor, we provide a smoothed *path* for the UAV that the drone can physically follow. Last, with the smoothed path, the time-optimal *trajectory* of the UAV flying through the path as fast as possible is determined. Three online algorithms are required to acquire the route, path and trajectory, and these three problems are much easier than the original problem to solve (shown below); therefore, they can all be efficiently executed in an online fashion.

A flow chart of the proposed methodological framework is illustrated in Fig. 4. In the remainder of this section, we explain the details of the methodological framework.

2.2. Control network

2.2.1. Network establishment

In the first (offline) stage, a control network must be established. The control network is invariant regardless of the real-time UAV request; therefore, it relates only to the existing terrains and buildings in the area of interest. For convenience, we treat each building as a cuboid; for non-cuboid-shaped buildings, constructing cuboids with appropriate sizes to contain them is a viable approach. As shown

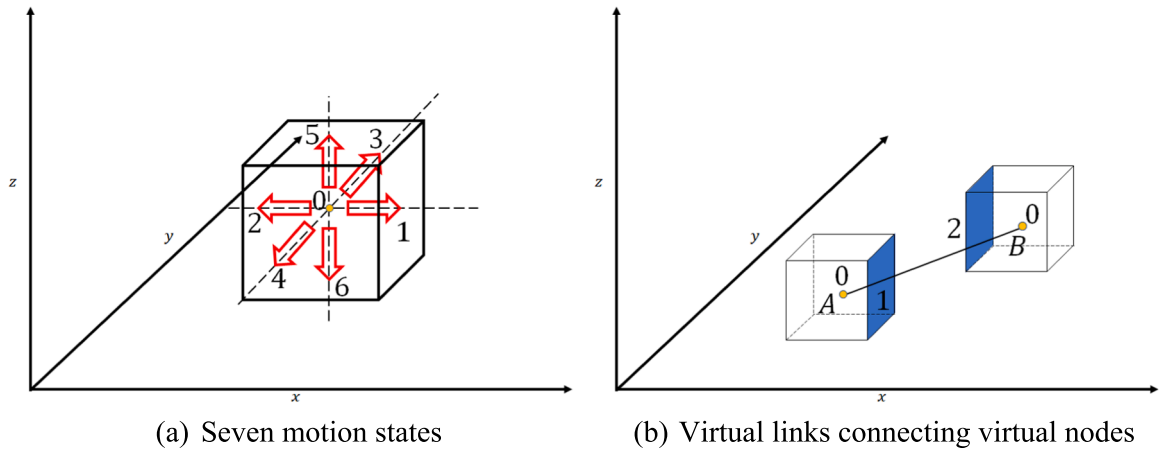


Fig. 8. Virtual nodes in a control point.

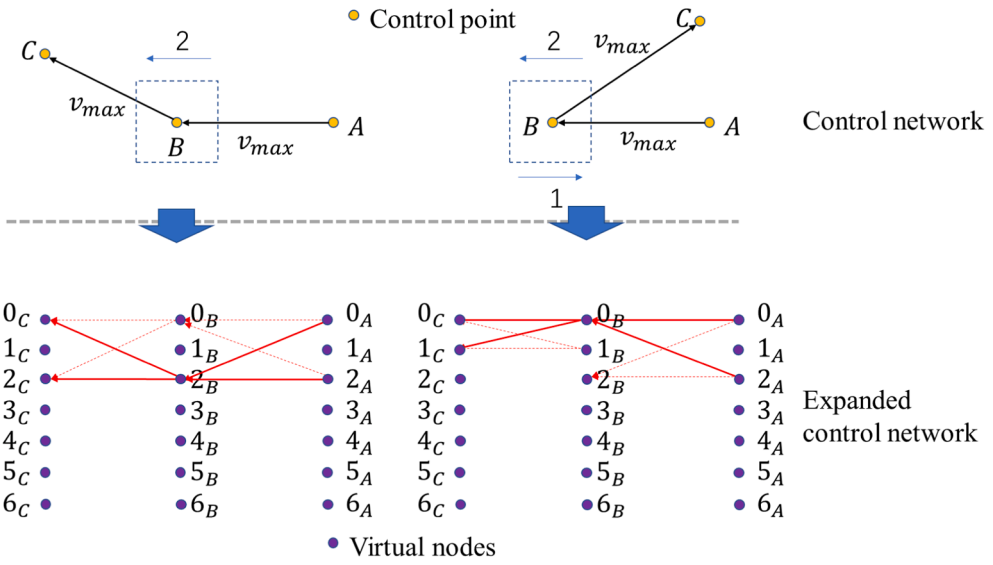


Fig. 9. Illustration of motion state feasibility.

in Fig. 5(a), the control points are set around the cuboid-shaped buildings (including one on the top). To ensure safety, each control point is at least d_1 meters away from the surfaces of the cuboid (projected) and at least d_2 meters above the ground. Additionally, we denote d_3 as the distance between two consecutive control points in the vertical direction, which is used as a measure of the density of control points. We can connect these control points with links to suggest a feasible flying sub path between two control points; the connection must ensure collision avoidance such that they cannot enter the inner space of a cuboid-shaped building. For each potential link, we conduct a conflict test to ensure that it will not collide with any building. As shown in Fig. 6, if a potential link passes through a building, then this link should not be connected for collision avoidance; and judging whether a link passes through a building is equivalent to judging whether it intersects with at least one surface of the building. Therefore, the process of the test is that: for each surface of each building, computing the coordinate of the intersection between the given link and the plane the surface lies in; if the intersecting point is within the range of building surface, then this potential link is not feasible and will not be connected. Additionally, to reduce the link sizes, we can set a maximum allowable distance d^* such that any pair of control points with distances larger than d^* is unconnected (Fig. 5(b)).

For a given network, the travel cost (or equivalently, flying time) on any link must be uniquely determined for calculating the optimal route in the online stage. However, as we will show below, the UAV flying time on the same link could be dependent on the subsequent link the UAV will pass through. For example, in the left-side case of control point B in Fig. 7, a UAV flies following the sequence $A \to B \to C$, and it experiences small-angle steering at B; with path smoothing, it is viable to fly through B at maximum speed. However, in the right-side case, a similar maneuver is not feasible because the UAV requires sharp steering at B, implying that it has to slow down at first and then accelerate. Therefore, for a UAV flying through the same link $A \to B$, different strategies can lead to

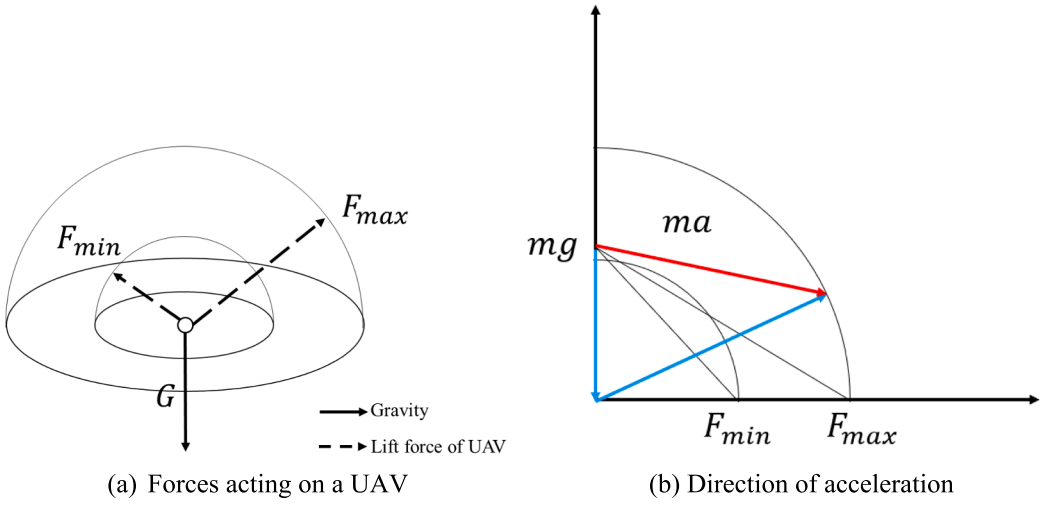


Fig. 10. Force analysis and acceleration.

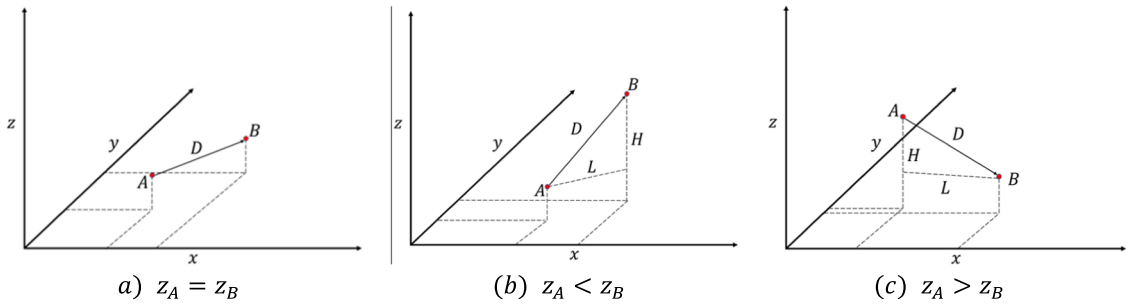


Fig. 11. Level flying, ascending, and descending (Node A and B represent an origin or a destination).

different link cost, which exerts an influence on the subsequent travel time evaluation. To capture this effect, we need to introduce virtual nodes at each control point to differentiate the UAV motion states.

In this study, we consider seven motion states for a UAV at each control point, as shown in Fig. 8(a). State 0 represents "holding", indicating that the UAV stops transiently around this control point. The other six states (1–6) indicate flying along six different directions with the associated maximum allowable speed, represented by the six surfaces of a cube in Fig. 8(a).

Next, the connection between these virtual nodes is considered. Notably, two virtual nodes belonging to two different control points are connected only if the two control points are connected by a link in the control network; for convenience, we call the network constituted by virtual nodes and their connections an *expanded control network*. As illustrated in Fig. 8(b), if the direction of link $A \rightarrow B$ belongs to state 1 (rightward), then we connect four pairs of virtual nodes: $(0_A \rightarrow 0_B), (0_A \rightarrow 1_B), (1_A \rightarrow 0_B), and (1_A \rightarrow 1_B)$. The meaning of $(0_A \rightarrow 0_B)$ is as follows: stop at A, accelerate, and decelerate to stop at B; the meaning of $(0_A \rightarrow 1_B)$ is as follows: stop at A and then accelerate to fly through B at the highest speed in direction 1; and the other two connections can be understood in similar ways. Obviously, for link $B \rightarrow A$ belonging to state 2 (leftward), we connect four pairs of virtual nodes: $(0_B \rightarrow 0_A), (0_B \rightarrow 2_A), (2_B \rightarrow 0_A), and (2_B \rightarrow 2_A)$.

Then, we explain why the above connection rule can maintain the feasibility issue illustrated in Fig. 7. As shown in Fig. 9, the expanded control networks are associated with the two cases drawn. For the left-side case, since both links $A \rightarrow B$ and $B \rightarrow C$ are oriented toward direction 2, in the expanded network representation, we see that the UAV can fly through B in state 2_B (the red solid lines), i.e., at full speed. However, for the right-side case, we observe that the UAV can fly only through B in state 0_B (the red solid lines), implying that it must first stop and then accelerate. Therefore, our proposed expanded network representation can effectively capture different motion states of the UAVs in different assigned routes. In the following section, the expanded control network is used in a computational procedure to obtain the optimal route, while the control network is primarily used for demonstration purposes.

Two issues require mentioning before explaining the next step. First, in some cases, the buildings are not sufficiently dense, which can lead to substantial detours with the control point setting given in Fig. 5(a). In this scenario, we can construct some virtual control points in the environment. Second, our virtual point setting is one possible method to alleviate the motion state feasibility issue. A more sophisticated setting may lead to a more accurate network description, but will increase the computational burden.

2.2.2. Link travel time evaluation

After constructing the (expanded) control network, we must evaluate the travel time on all links to provide guidance for optimal route finding as well as other advanced applications. We use simple dynamics and UAV kinetic constraints to derive our results. Fig. 10 illustrates the analysis of UAV kinetic properties. As shown in Fig. 10(a), there are two forces acting on the UAV: gravity, $\boldsymbol{G} = m\boldsymbol{g}$, and lift force, $\boldsymbol{F}$. For a UAV, we can reasonably assume that the direction of $\boldsymbol{F}$ can be only upward, and it is within the range of $|\boldsymbol{F}| \in [F_{min}, F_{max}]$; therefore, the feasible region of the lift force constitutes an upper spherical shell (Fig. 10(a)). When the flying speed is not too high, the drag force of a UAV is generally much smaller compared to the lift force (e.g., with a speed of 15m/s, the drag force of an ordinary-shape UAV is generally no more than $0.5N$, as shown in Felismina et al. (2017)), so in this section we neglect the impact of drag force.

By simple force analysis, we can derive the analytical expressions for the maximum acceleration in different directions (Fig. 10(b)). In the cost evaluation, we always set the acceleration (and deceleration) to be as large as possible to obtain the minimum flying time. A straight-line flying task from A to B is shown in Fig. 11. The heights of points A and B are denoted by z_A and z_B, respectively. The flying distance is denoted by D, the horizontal distance is denoted by L, and the vertical distance is denoted by H ($H \geq 0$ in any situation). We use $\boldsymbol{F}_{AB}$ and $\boldsymbol{F}_{BA}$ to denote the maximum lift forces for acceleration and deceleration in directions A and B, and $\boldsymbol{a}_{AB}$ and $\boldsymbol{a}_{BA}$ to denote the associated acceleration and deceleration, respectively. Through a case-by-case analysis, we obtain the following results:

$$|\boldsymbol{F}_{AB}| = F_{max},\ |\boldsymbol{a}_{AB}| = \sqrt{\frac{F_{AB}^2}{m^2} - \frac{g^2L^2}{D^2}} - \frac{|g|H}{D} \text{ if } z_A \leq z_B$$

$$|\boldsymbol{F}_{AB}| = F_{max},\ |\boldsymbol{a}_{AB}| = \sqrt{\frac{F_{AB}^2}{m^2} - \frac{g^2L^2}{D^2}} + \frac{|g|H}{D} \text{ if } z_A > z_B,$$

$$\frac{H}{D} \in \left(0, \frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{max}^2}}\right]$$

$$|\boldsymbol{F}_{AB}| = \frac{|\boldsymbol{G}|L}{H},\ |\boldsymbol{a}_{AB}| = \sqrt{\frac{F_{AB}^2}{m^2} + \boldsymbol{g}^2} \text{ if } z_A > z_B,$$

$$\frac{H}{D} \in \left(\frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{max}^2}}, \frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{min}^2}}\right]$$

$$|\boldsymbol{F}_{AB}| = F_{min},\ |\boldsymbol{a}_{AB}| = \frac{|g|H}{D} - \sqrt{\frac{F_{AB}^2}{m^2} - \frac{g^2L^2}{D^2}} \text{ if } z_A > z_B,$$

$$\frac{H}{D} \in \left(\frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{min}^2}}, 1\right]$$

$$|\boldsymbol{F}_{BA}| = F_{max},\ |\boldsymbol{a}_{BA}| = \sqrt{\frac{F_{BA}^2}{m^2} - \frac{g^2L^2}{D^2}} - \frac{|g|H}{D} \text{ if } z_A \geq z_B$$

$$|\boldsymbol{F}_{BA}| = F_{max},\ |\boldsymbol{a}_{BA}| = \sqrt{\frac{F_{BA}^2}{m^2} - \frac{g^2L^2}{D^2}} + \frac{|g|H}{D} \text{ if } z_A < z_B,$$

$$\frac{H}{D} \in \left(0, \frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{max}^2}}\right]$$

$$|\boldsymbol{F}_{BA}| = \frac{|\boldsymbol{G}|L}{H},\ |\boldsymbol{a}_{BA}| = \sqrt{\frac{F_{BA}^2}{m^2} + \boldsymbol{g}^2} \text{ If } z_A < z_B,$$

$$\frac{H}{D} \in \left(\frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{max}^2}}, \frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{min}^2}}\right]$$

$$|\boldsymbol{F}_{BA}| = F_{min},\ |\boldsymbol{a}_{BA}| = \frac{|g|H}{D} - \sqrt{\frac{F_{BA}^2}{m^2} - \frac{g^2L^2}{D^2}} \text{ if } z_A < z_B,$$

$$\frac{H}{D} \in \left(\frac{|\boldsymbol{G}|}{\sqrt{\boldsymbol{G}^2 + \boldsymbol{F}_{min}^2}}, 1\right]$$

Then, we discuss the restrictions on UAV cruising speed. For security reasons, a maximum allowable flying speed, v_{max}/ a maximum allowable ascending speed, v_{up}; and a maximum allowable descending speed, v_{down}, are set. Therefore, denoting v_u as the maximum speed in the direction from A to B, we have

$$v_u = v_{max} \text{ if } z_A = z_B$$

$$v_u = \min\left\{v_{max}, \frac{v_{up}D}{H}\right\} \text{ if } z_A < z_B$$

$$v_u = \min\left\{v_{max}, \frac{v_{down}D}{H}\right\} \text{ if } z_A > z_B$$

Now, we present the link travel time evaluation. A pair of control points is connected by 4 virtual links in the expanded control network, each with different initial and final motion state. Therefore, we also need to consider four scenarios in evaluating the link travel time. We denote s_A and s_B as the motion state of the UAV at A and B, respectively, where $s_A = 0$ indicates the holding state at A according to Fig. 8(a), and $s_A \neq 0$ indicates the state at maximum speed in the associated direction. If $s_A = 0$ and $s_B = 0$, then we have

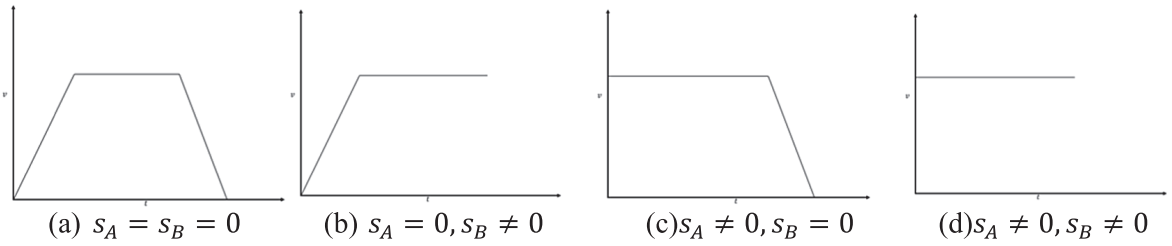


Fig. 12. Speed curves on different virtual links.

$$t_{AB} = \begin{cases} \dfrac{v_u}{a_{AB}} + \dfrac{v_u}{a_{BA}} + \dfrac{D - \left(\dfrac{1}{a_{AB}} + \dfrac{1}{a_{BA}}\right)\dfrac{v_u^2}{2}}{v_u} & D \geq \left(\dfrac{1}{a_{AB}} + \dfrac{1}{a_{BA}}\right)\dfrac{v_u^2}{2} \\ \sqrt{2D\left(\dfrac{1}{a_{AB}} + \dfrac{1}{a_{BA}}\right)} & D < \left(\dfrac{1}{a_{AB}} + \dfrac{1}{a_{BA}}\right)\dfrac{v_u^2}{2} \end{cases} \tag{1}$$

If $s_A = 0$ and $s_B \neq 0$, then we have

$$t_{AB} = \begin{cases} \dfrac{v_u}{a_{AB}} + \dfrac{D - \dfrac{v_u^2}{2a_{AB}}}{v_u} & D \geq \dfrac{v_u^2}{2a_{AB}} \\ \sqrt{\dfrac{2D}{a_{AB}}} & D < \dfrac{v_u^2}{2a_{AB}} \end{cases} \tag{2}$$

If $s_A \neq 0$ and $s_B = 0$, then we have

$$t_{AB} = \begin{cases} \dfrac{v_u}{a_{BA}} + \dfrac{D - \dfrac{v_u^2}{2a_{BA}}}{v_u} & D \geq \dfrac{v_u^2}{2a_{BA}} \\ \sqrt{\dfrac{2D}{a_{BA}}} & D < \dfrac{v_u^2}{2a_{BA}} \end{cases} \tag{3}$$

Finally, if $s_A \neq 0$ and $s_B \neq 0$, then we have

$$t_{AB} = \frac{D}{v_u} \tag{4}$$

The speed curves corresponding to the above four scenarios are given in Fig. 12.

2.3. Route finding and 3D Dubins curve smoothing

With the control network and its expansion established in Section 2.2, we are now ready to proceed with the details in the second (online) stage. For the online stage, a UAV flying task is presented with a given origin and destination. Then, with the established extended control network, we can identify one *route* with the shortest total travel time evaluated by the link costs provided in Section 2.2. The optimal route determination can be executed with Dijkstra's algorithm (Biyela and Rawatlal, 2019), which is efficient and easy to implement.

As introduced in Section 2.1, an optimal route essentially provides a collision-free "corridor" for the UAV, and we need to provide a smooth path for the UAV within the corridor. We adopt a 3D Dubins curve smoothing technique (Dubins, 1957; Stefano et al., 2018) for this purpose. A route in the control network is represented by a set of consecutive control points $\{\boldsymbol{p}_0, \boldsymbol{p}_1, \boldsymbol{p}_2, \cdots, \boldsymbol{p}_m\}$. Within the route, any three consecutive control points $\{\boldsymbol{p}_i, \boldsymbol{p}_{i+1}, \boldsymbol{p}_{i+2}\}$ can determine a 2D plane in 3D space. We first transform the 3D coordinate into a 2D coordinate and project the control points onto it. We generate a pose matrix with a size of 4×4 and in the form $\begin{pmatrix} \boldsymbol{R} & \boldsymbol{P} \\ 0 & 1 \end{pmatrix}$, where $\boldsymbol{R}$ is a 3×3 orientation rotation matrix, $\boldsymbol{P}$ is a 3×1 position vector, and $\boldsymbol{P}$ defaults to $(0,0,0)^{\mathrm{T}}$. Then, we perform the following transformation:

$$\begin{pmatrix} \boldsymbol{R} & \boldsymbol{P} \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} p_{i,x} & p_{i,y} & p_{i,z} & 1 \\ p_{i+1,x} & p_{i+1,y} & p_{i+1,z} & 1 \\ p_{i+2,x} & p_{i+2,y} & p_{i+2,z} & 1 \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} \widehat{p}_{i,x} & \widehat{p}_{i,y} & 0 & 1 \\ \widehat{p}_{i+1,x} & \widehat{p}_{i+1,y} & 0 & 1 \\ \widehat{p}_{i+2,x} & \widehat{p}_{i+2,y} & 0 & 1 \end{pmatrix}^{\mathrm{T}} \tag{5}$$

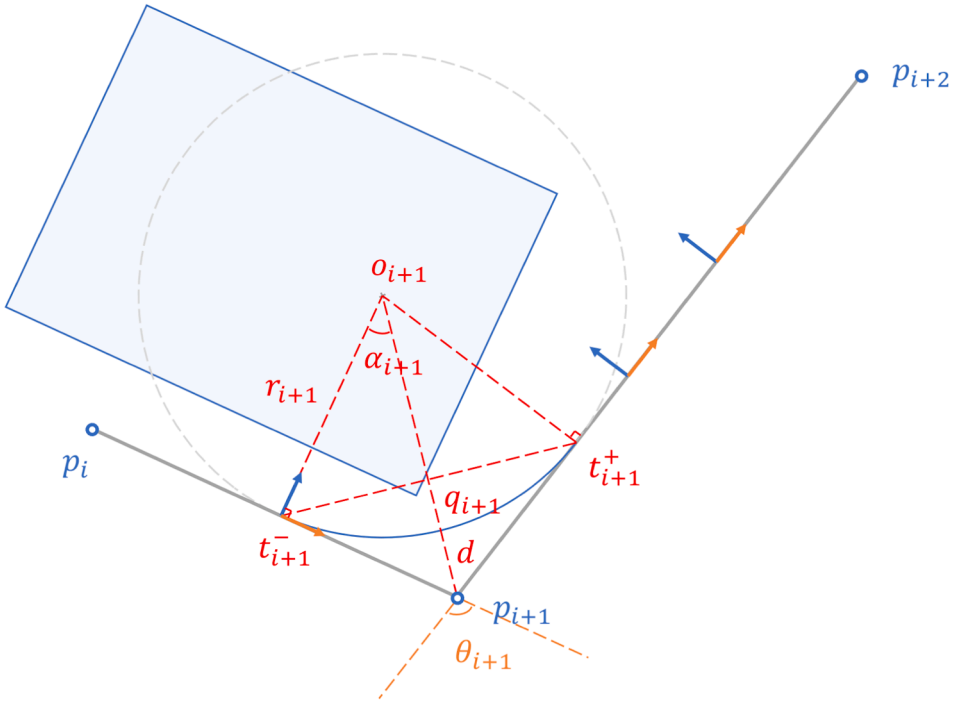


Fig. 13. Illustration of curve smoothing.

The coordinates of $\boldsymbol{p}_i, \boldsymbol{p}_{i+1}, \boldsymbol{p}_{i+2}$ on the 2D plane are then given by $\widehat{\boldsymbol{p}}_i = \left(\widehat{p}_{i,x}, \widehat{p}_{i,y}\right), \widehat{\boldsymbol{p}}_{i+1} = \left(\widehat{p}_{i+1,x}, \widehat{p}_{i+1,y}\right), \widehat{\boldsymbol{p}}_{i+2} = \left(\widehat{p}_{i+2,x}, \widehat{p}_{i+2,y}\right)$. This equation has an infinite number of solutions, but combined with the restrictions on the orientation rotation matrix, $\boldsymbol{R}$, we can acquire a unique solution to $\widehat{\boldsymbol{p}}_i$, $\widehat{\boldsymbol{p}}_{i+1}$, and $\widehat{\boldsymbol{p}}_{i+2}$.

Now, we aim to smooth the flying path around control point p_{i+1}, as shown in Fig. 13. We set the deviation of the path to the control point as d, and the quantity of d ensures that the path does not intersect with the building (a certain safety distance is required). With d, we can uniquely determine the smoothened steering curve with center $\boldsymbol{o}_{i+1}$, radius r_{i+1} and the starting and ending points of steering, i.e., $\boldsymbol{t}^-_{i+1}$ and $\boldsymbol{t}^+_{i+1}$. The angle formed by $\boldsymbol{p}_i, \boldsymbol{p}_{i+1}, \boldsymbol{p}_{i+2}$ is θ_{i+1}. Then, we can compute the above parameters by the following equations:

$$\frac{r_{i+1}}{r_{i+1}+d} = \sin\frac{\theta_{i+1}}{2} \tag{6}$$

$$\left|\boldsymbol{p}_{i+1}, \boldsymbol{t}^-_{i+1}\right| = \left|\boldsymbol{p}_{i+1}, \boldsymbol{t}^+_{i+1}\right| = \frac{r_{i+1}}{\tan\frac{\theta_{i+1}}{2}} \tag{7}$$

where $|\cdot|$ is a distance operator. Then, on the transformed 2D plane, the center of the steering curve $\widehat{\boldsymbol{o}}_{i+1} = \left(\widehat{o}_{i+1,x}, \widehat{o}_{i+1,y}\right)$ is derived by

$$q = \sqrt{\left(\widehat{t}^+_{i+1,x} - \widehat{t}^-_{i+1,x}\right)^2 + \left(\widehat{t}^+_{i+1,y} - \widehat{t}^-_{i+1,y}\right)^2} \tag{8}$$

$$\widehat{o}_{i+1,x} = \frac{\left(\widehat{t}^+_{i+1,x} + \widehat{t}^-_{i+1,x}\right)}{2} \pm \sqrt{\left(r^2_{i+1} - \frac{q^2}{4}\right)} \frac{\left(\widehat{t}^+_{i+1,y} - \widehat{t}^-_{i+1,y}\right)}{q} \tag{9}$$

$$\widehat{o}_{i+1,y} = \frac{\left(\widehat{t}^+_{i+1,y} + \widehat{t}^-_{i+1,y}\right)}{2} \pm \sqrt{\left(r^2_{i+1} - \frac{q^2}{4}\right)} \frac{\left(\widehat{t}^+_{i+1,x} - \widehat{t}^-_{i+1,x}\right)}{q} \tag{10}$$

where the choice of signs in Eqs. (9)-(10) requires careful treatment by case. With the above formulae, we can acquire the expression for the steering curve in the 2D plane and transform the curve back to the original 3D space to yield the smoothened path around control point $\boldsymbol{p}_{i+1}$.

With the aforementioned smoothing technique, we can proceed from $\{\boldsymbol{p}_0, \boldsymbol{p}_1, \boldsymbol{p}_2\}$ to $\{\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3\}$, $\{\boldsymbol{p}_2, \boldsymbol{p}_3, \boldsymbol{p}_4\}$, … and to $\{\boldsymbol{p}_{m-2}, \boldsymbol{p}_{m-1}, \boldsymbol{p}_m\}$ to acquire the whole smoothened path for the UAV from origin to destination. However, this maneuver can lead to some infeasible

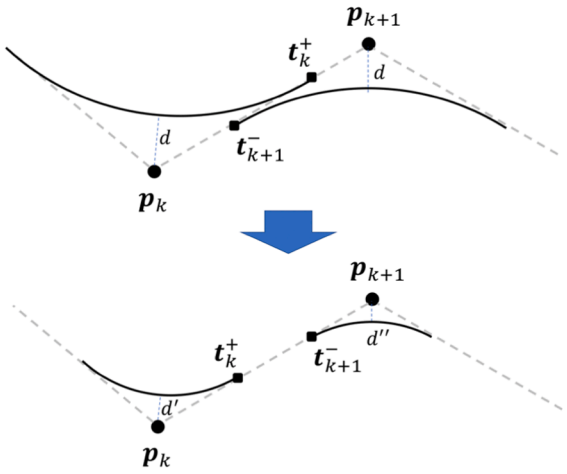


Fig. 14. Infeasible smoothened path and its fixation.

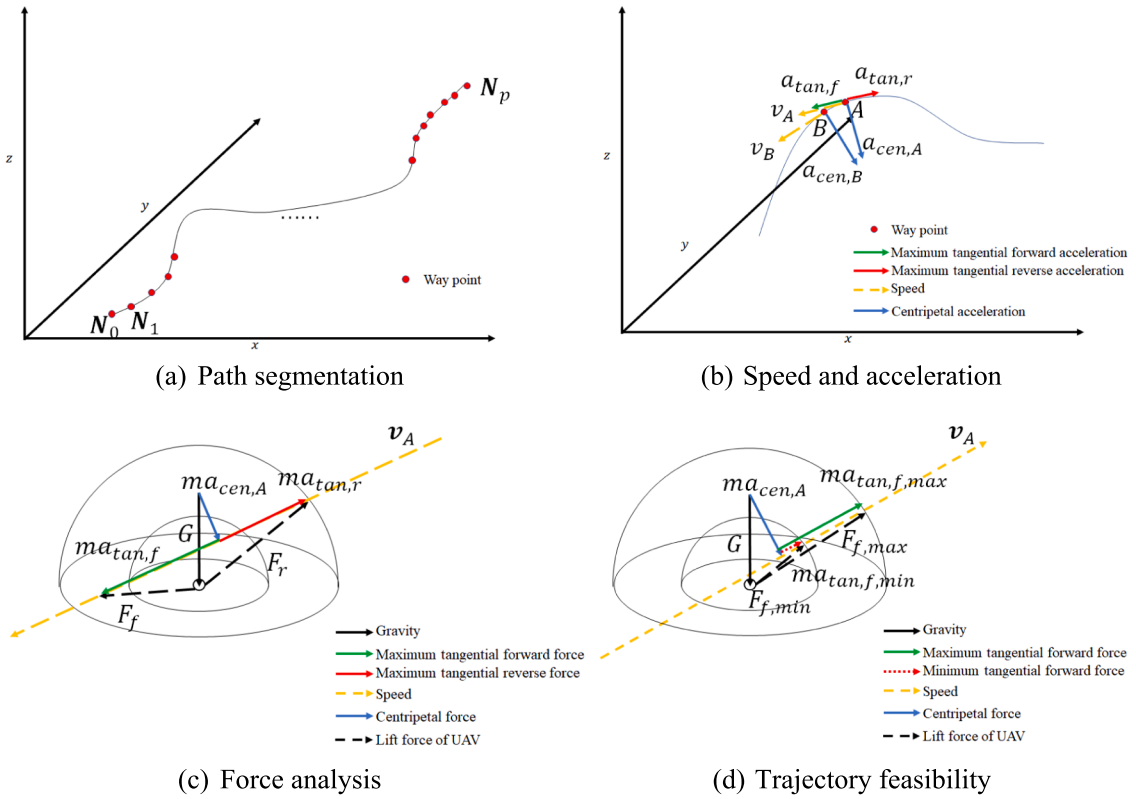


Fig. 15. Illustrations of time-optimal trajectory generation.

cases. For example, two consecutive control points p_k and p_{k+1} are spatially very close to each other. When the steering curve around p_k is determined, it may not be physically possible to deviate the curve a distance of d around p_{k+1} (see the upper half of Fig. 14). When encountering this case, we reduce the curve deviations associated with those two control points. For example, as shown in the lower half of Fig. 14, we shrink the deviation of p_k to d' and that of p_{k+1} to d'', and then, the feasibility of the smoothed path is maintained. This fix will be used until the whole smooth path is feasible.

2.4. Time-Optimal trajectory generation

In the last subsection of the methodological section, we acquire a time-optimal space–time *trajectory* based on the smoothed path

provided in Section 3.3. Given a spatial path, finding the time-optimal speed profile and acceleration plan is not a difficult task. In this study, we adopted a modified version of an existing algorithm (Hoffmann et al. 2008) to solve this problem.

A series of nodes $\{N_o, N_1, \cdots, N_p\}$ sliced the spatial path into segments, as shown in Fig. 15(a). For node i, we can approximately calculate the radius of curvature r_i. Then, as shown in Fig. 15(b), the directions of speed, tangential acceleration, and centripetal acceleration in each node can be determined, and the centripetal acceleration of two consecutive way points A and B are obtained by

$$|\boldsymbol{a}_{cen,A}| = \frac{v_A^2}{r_A}$$

$$|\boldsymbol{a}_{cen,B}| = \frac{v_B^2}{r_B}$$

If the velocity at point A, i.e., v_A, is given, as shown in Fig. 15(c), based on the force analysis shown in Fig. 10, we can obtain

$$\boldsymbol{G} + \boldsymbol{F}_f = m\boldsymbol{a}_{cen,A} + m\boldsymbol{a}_{tan,f}$$

$$\boldsymbol{G} + \boldsymbol{F}_r = m\boldsymbol{a}_{cen,A} + m\boldsymbol{a}_{tan,r}$$

where $\boldsymbol{F}_f$ and $\boldsymbol{F}_r$ denote the forward and reverse forces, respectively. Then, we can calculate the maximum tangential forward acceleration $a_{tan,f}$ and the maximum tangential reverse acceleration $a_{tan,r}$, with the process graphically shown in Fig. 15(c). If the segmentation is sufficiently accurate, the direction vectors of tangential acceleration $a_{tan,f}$ and $a_{tan,r}$ can be regarded as constant vectors from point A to successive point B.

Following the aforementioned analyses, trajectory generation can be executed in a forward manner, i.e., from origin to destination, and at each step, the UAV myopically accelerates to generate as much speed as fast as possible. Similar to path smoothing, during the process of trajectory generation, some feasibility issues exist. For example, as shown in Fig. 15(d), the UAV can increase speed only when the net force provides sufficient centripetal acceleration, and the trajectory is infeasible when the maximum allowable speed at B is smaller than that at A. To alleviate this issue, we need to design a "backward tracing" mechanism such that the system will trace several steps back to decelerate earlier to ensure the speed feasibility at the current node.

The detailed procedure for time-optimal trajectory generation is illustrated below. In the procedure, d_{AB} is used to denote the distance between A and B. The "backward tracing" mechanism is depicted in Step 3 of the algorithm.

Time-optimal trajectory generation algorithm

Step 0. Initialize $z = 0, v_{a,z} = 0, T_z = 0, \boldsymbol{A}_z = \boldsymbol{N}_o, \boldsymbol{B}_z = \boldsymbol{N}_1$, where, $\boldsymbol{A}_z$ and $\boldsymbol{B}_z$ is the origin and destination of iteration z, $v_{a,z}$ is the speed at the point $\boldsymbol{A}_z$ and T_z is the total travel time from $\boldsymbol{N}_o$ to $\boldsymbol{N}_z$.

Step 1. Calculate the centripetal acceleration $a_{cen,A} = \frac{v_{a,z}^2}{r_a}$, and the maximum tangential forward and reverse acceleration $a_{tan,f} = \sqrt{|\boldsymbol{g} + \frac{\boldsymbol{F}_f}{m}|^2 - a_{cen,A}^2}, a_{tan,r} = \sqrt{|\boldsymbol{g} + \frac{\boldsymbol{F}_r}{m}|^2 - a_{cen,A}^2}$, $\boldsymbol{g} + \frac{\boldsymbol{F}_f}{m}$ and $\boldsymbol{g} + \frac{\boldsymbol{F}_r}{m}$ are the maximum acceleration that can provide enough centripetal acceleration for tangential forward and reverse acceleration calculated through Fig. 15

Step 2. Calculate the maximum speed $v_{b,z} = \min\{v_{1,z}, v_{2,z}, v_{3,z}\}$ at the point $\boldsymbol{B}_z$, where $v_{1,z}$ is the speed limit of UAV, $|v_{1,z}| \le v_{max}, \theta$ is the flight angle,

$$|v_{1,z}|sin\theta \le \begin{cases} v_{up}, \theta \ge 0 \\ v_{down}, \theta < 0 \end{cases}$$

$v_{2,z}$ is the maximum speed based on centripetal acceleration, $v_{2,z} = \sqrt{a_b r_b}$, a_b is the maximum acceleration at point $\boldsymbol{B}_z$ calculated through method in Section 2.2, r_b is the radius of curvature;

$v_{3,z}$ is an additional limitation of speed, with an initialization of $v_{3,z} = v_{max}$.

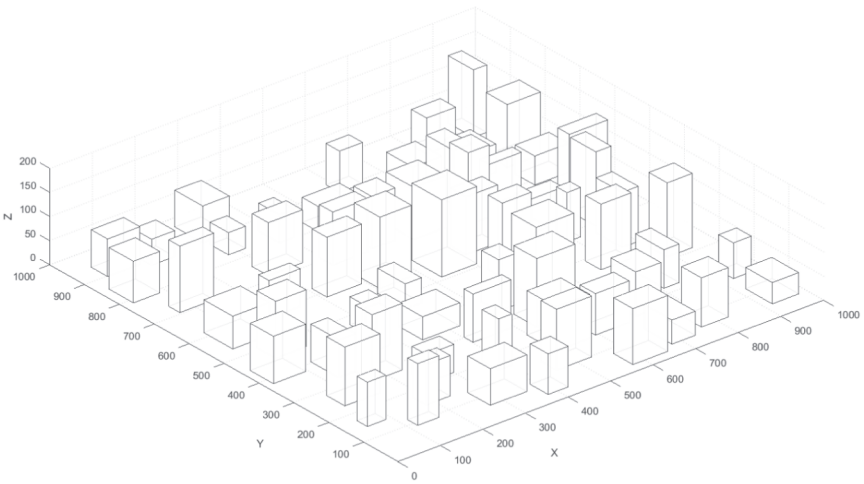
Step 3. If $v_{a,z} = v_{b,z}$, let $T_z = T_{z-1} + \frac{d_{AB}}{v_{a,z}}$, $z = z+1, \boldsymbol{A}_z = \boldsymbol{N}_z$; if $\boldsymbol{A}_z = \boldsymbol{N}_p$, output T_z and stop, otherwise set $\boldsymbol{B}_z = \boldsymbol{N}_{z+1}$, $v_{a,z} = v_{a,z-1}$ and go to Step 1;

If $v_{a,z} > v_{b,z}$ and $d_{AB} < \frac{v_{a,z}^2 - v_{b,z}^2}{2a_{tan,r}}$, then let $v_{3,z-1} = \sqrt{2a_{tan,r}d_{AB} + v_{b,z}^2}$, $z = z-1$ and go to Step 1;

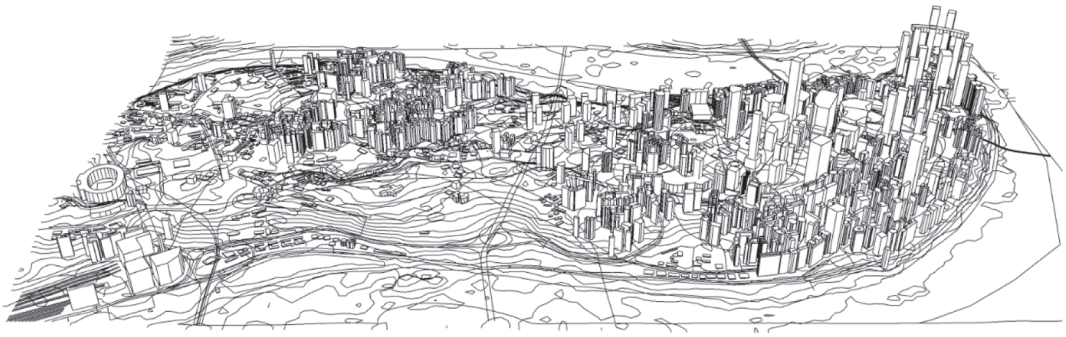
If $v_{a,z} > v_{b,z}$ and $d_{AB} \ge \frac{v_{a,z}^2 - v_{b,z}^2}{2a_{tan,r}}$, then let $T_z = T_{z-1} + \frac{v_{a,z} - v_{b,z}}{a_{tan,r}} + \frac{d_{AB} - \frac{v_{a,z}^2 - v_{b,z}^2}{2a_{tan,r}}}{v_{a,z}}$, $z = z+1, \boldsymbol{A}_z = \boldsymbol{N}_z$; if $\boldsymbol{A}_z = \boldsymbol{N}_p$, output T_z and stop, otherwise $\boldsymbol{B}_z = \boldsymbol{N}_{z+1}$, $v_{a,z} = v_{b,z-1}$ and go to Step 1;

If $v_{a,z} \le v_{b,z}$ and $d_{AB} < \frac{v_{b,z}^2 - v_{a,z}^2}{2a_{tan,f}}$, then let $T_z = T_{z-1} + \frac{\sqrt{2a_{tan,r}d_{AB} + v_{a,z}^2} - v_{a,z}}{a_{tan,f}}$, $z = z+1, \boldsymbol{A}_z = \boldsymbol{N}_z$; if $\boldsymbol{A}_z = \boldsymbol{N}_p$, output T_z and stop, otherwise $\boldsymbol{B}_z = \boldsymbol{N}_{z+1}$, $v_{a,z} = \sqrt{2a_{tan,r}d_{AB} + v_{a,z-1}^2}$ and go to Step 1;

If $v_{a,z} \le v_{b,z}$ and $d_{AB} \ge \frac{v_{b,z}^2 - v_{a,z}^2}{2a_{tan,f}}$, then $T_z = T_{z-1} + \frac{v_{b,z} - v_{a,z}}{a_{tan,f}} + \frac{d_{AB} - \frac{v_{b,z}^2 - v_{a,z}^2}{2a_{tan,f}}}{v_{b,z}}$, $z = z+1, \boldsymbol{A}_z = \boldsymbol{N}_z$; if $\boldsymbol{A}_z = \boldsymbol{N}_p$, output T_z and stop, otherwise $\boldsymbol{B}_z = \boldsymbol{N}_{z+1}$, $v_{a,z} = v_{b,z-1}$ and go to Step 1.



(a) The virtual city



(b) Yuzhong Peninsula in Chongqing, China

Fig. 16. Graphical illustration of two test areas.

3. Numerical experiments

3.1. Experiment settings

The proposed methodological framework was tested with computer simulation experiments. In our experiments, we used MATLAB R2020a and JAVA 1.8 based on the AMD computing platform with 16G GB of RAM and 1.7 GHz AMD Ryzen 4750u CPU. Two urban test areas (a virtual one and a real-world one) were established, as shown in Fig. 16. In the virtual city, we arranged 67 densely distributed buildings in an area of 1000m $\times$ 1000m, with the lengths and widths of the buildings ranging from 30 to 60 m and their heights ranging from 50 to 160 m, and the control points were selected according to the rules stated in Section 2.2 with $d^* = 300$m and $d_3 = 10$m. On the other hand, the real-world city was chosen to be an approximately 5000m $\times$ 2000m zone in Chongqing, a south-western Chinese city with complex terrain and multiple skyscrapers of heights ranging from 200 to 400 m. Overviews of the two test areas are shown in Fig. 16. Throughout the experiments, we set $v_{max} = 15m/s$, $v_{up} = 6m/s$, $v_{down} = 2m/s$, $m = 1kg$, $F_{max} = 16N$, and $F_{min} = 9N$.

In the virtual city tests, we implemented a benchmark method based on particle swarm optimization (PSO) as a comparison (Kennedy and Eberhart, 2002). Adopting PSO for trajectory planning requires discretizing the time horizon into a large number of steps; at each time step, a large number of particles are generated, and a cost function is defined to evaluate the performance of each particle. The cost function incorporates the consideration of destination reachability, total flying time, and collision avoidance along the path. In a complicated urban environment, the potential choices of paths are extensive, and we alleviate this issue by proposing a three-phase scheme as follows:

Step 1: Using PSO algorithm to find a feasible spatial path for the UAV.

Step 2: Smoothen the obtained spatial path with Dubins curve method.

Step 3: With the given spatial path, find the time-optimal trajectory with the method proposed in Section 2.4.

3.2. Tests based on a virtual city

In the virtual city, we chose a total of five origin–destination (OD) pairs to illustrate our results. The expanded control network contains a total of 33,572 nodes and 7,366,878 links. The overviews of the planned paths in the city for these five OD pairs are presented as follows:

1) OD pair 1

The planned path for OD pair 1 is shown in Fig. 17a. We observe that the acquired path is gradually ascending from the origin to the destination, and it flies over some low-rise buildings while making detours to bypass some tall buildings. The total flying time obtained by the time-optimal trajectory generation is 70.03 s.

2) OD pair 2

The planned path for OD pair 2 is shown in Fig. 17b. We observe that the acquired path gradually ascends first and then quickly descends, and the whole path makes a considerable detour. The total flying time obtained by the time-optimal trajectory generation is 85.95 s.

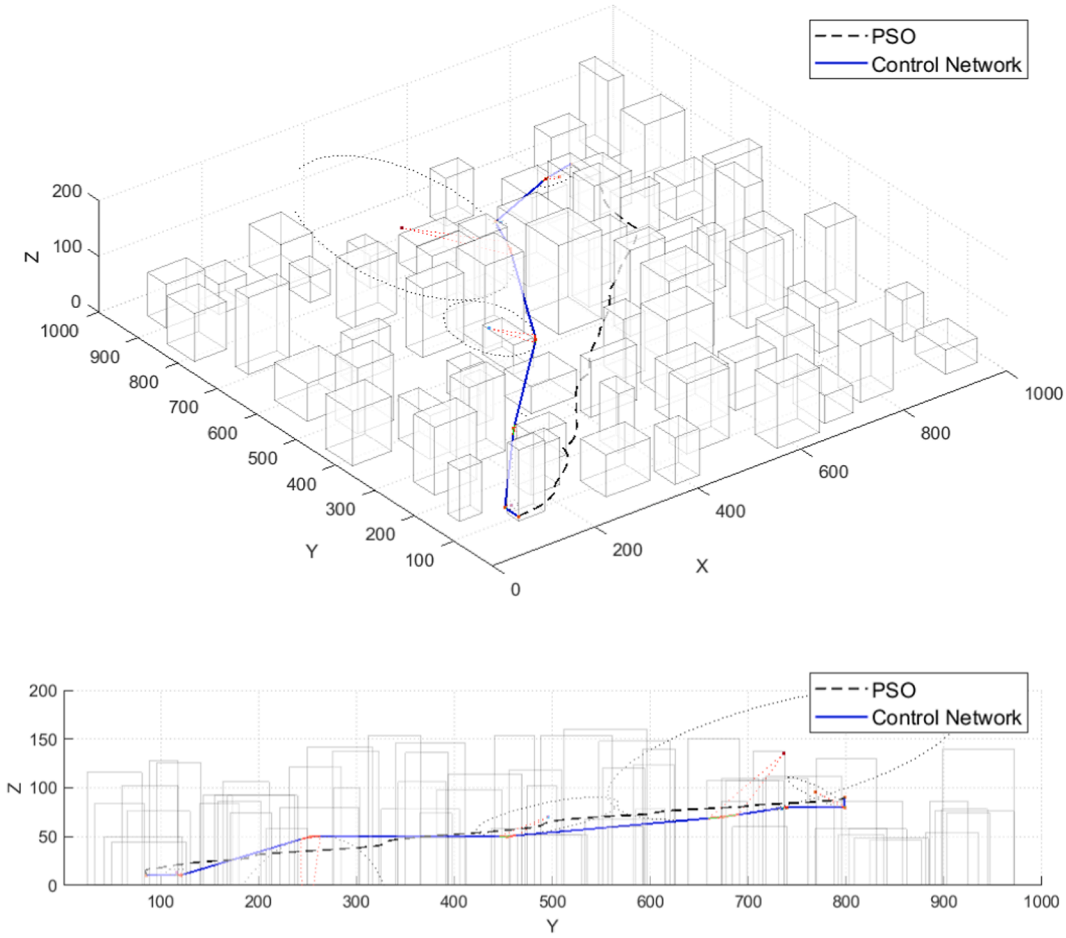


Fig. 17a. Acquired paths for OD pair 1.

3) OD pair 3

The planned path for OD pair 3 is shown in Fig. 17c. Similar to the case of OD pair 2, the whole path makes a considerable detour to obviate densely tall building groups. The total flying time obtained by the time-optimal trajectory generation is 65.02 s.

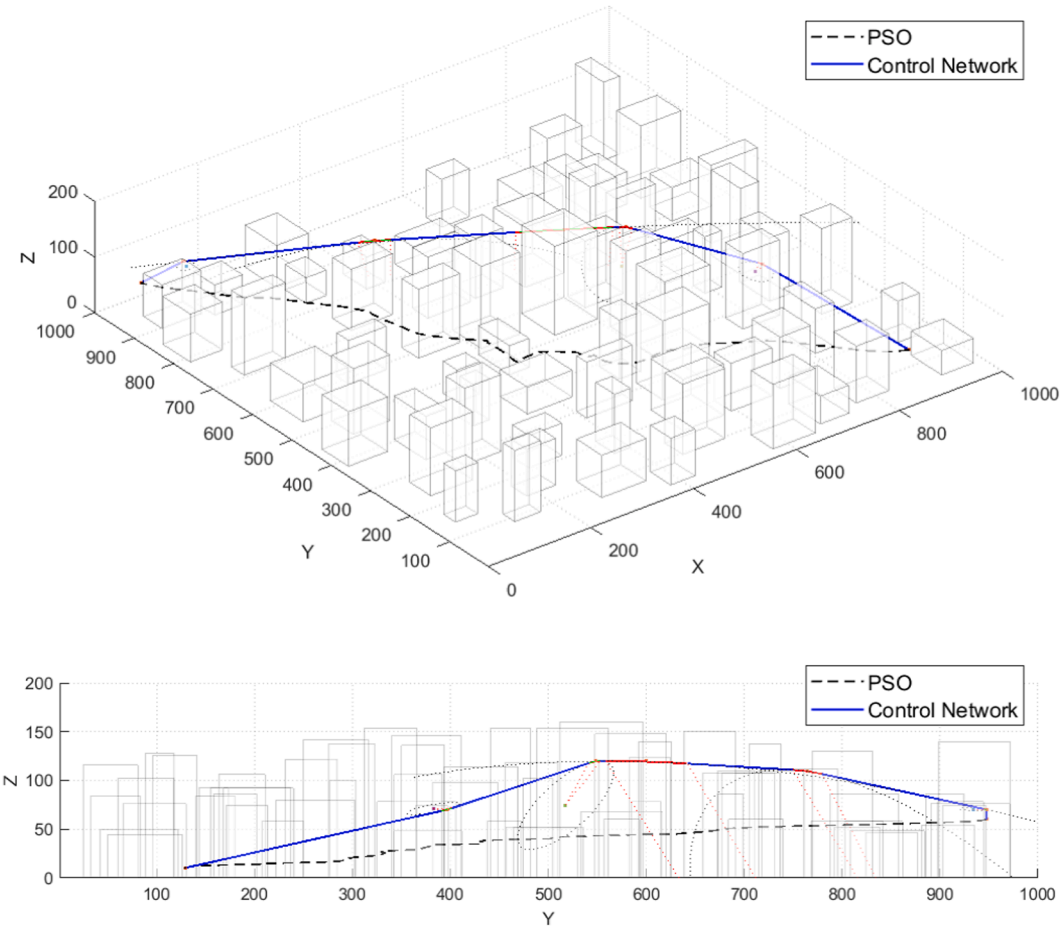


Fig. 17b. Acquired paths for OD pair 2.

4) OD pair 4

The planned path for OD pair 4 is shown in Fig. 17d. In contrast to the previous two cases, the planned path has a zigzag shape to constantly bypass buildings. The total flying time obtained by the time-optimal trajectory generation is 61.58 s.

5) OD pair 5

The planned path for OD pair 5 is shown in Fig. 17e. In this case, the planned path is quite close to a straight line crossing through the gaps between two neighboring tall buildings. The total flying time obtained by the time-optimal trajectory generation is 52.84 s.

Table 1 compares the control network approach with the benchmark PSO approach, focusing on the obtained flying time and the computational effort required. It was quickly determined that the online computing times for the control network approach were all less than 1 s on a personal computer, indicating that our proposed approach can satisfy real-time computing requirements; as a comparison, the PSO approach requires nearly 1,000 s for acquiring the result, implying that the control network approach improves the computational efficiency with more than 1,000 times. What's more, we observe that the flying times acquired by the control network approach are all smaller than those acquired by PSO, with a reduction of 1.61%-12.66%. The results shown in Table 1 demonstrate the superiority of the proposed approach in both solution quality and efficiency.

To better understand why the control network approach acquires better trajectories than PSO approach, we plot the associated speed-time curves for OD pair 5 in Fig. 18. As demonstrated, although the path acquired by PSO may has shorter distance (see Fig. 17e),

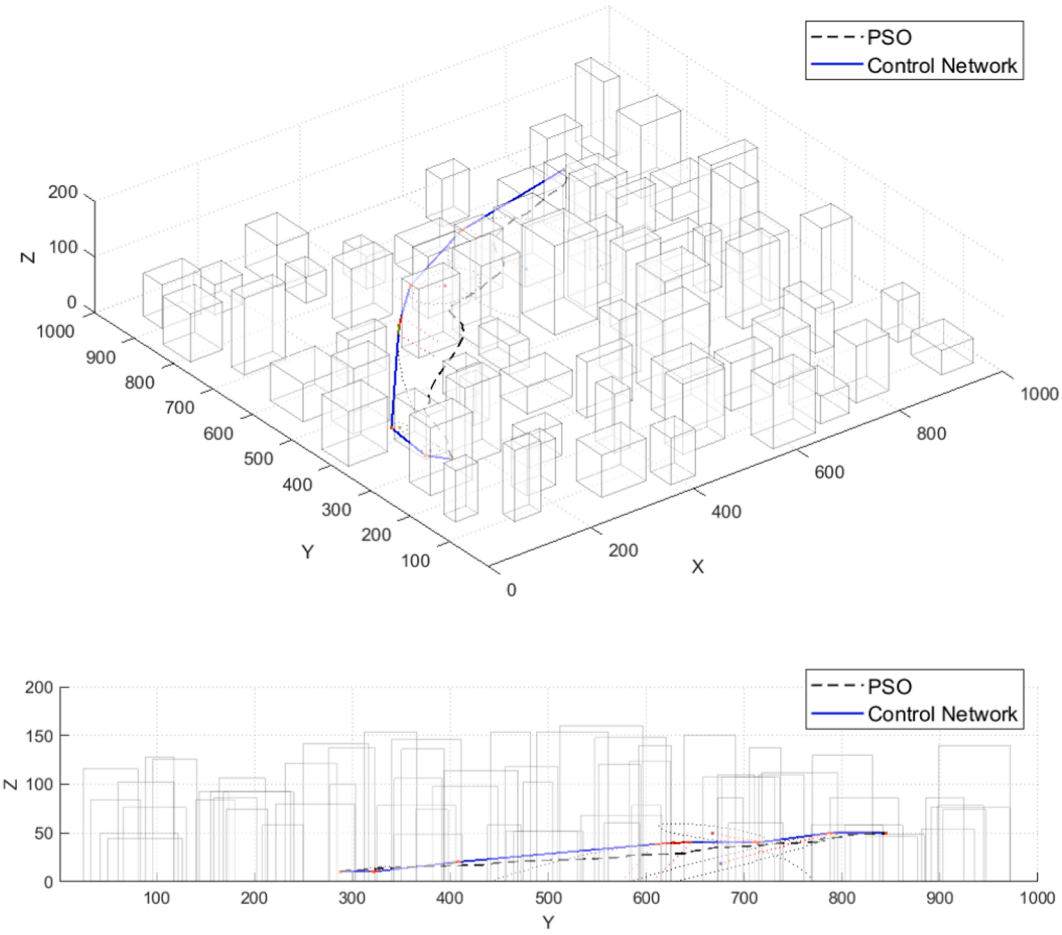


Fig. 17c. Acquired paths for OD pair 3.

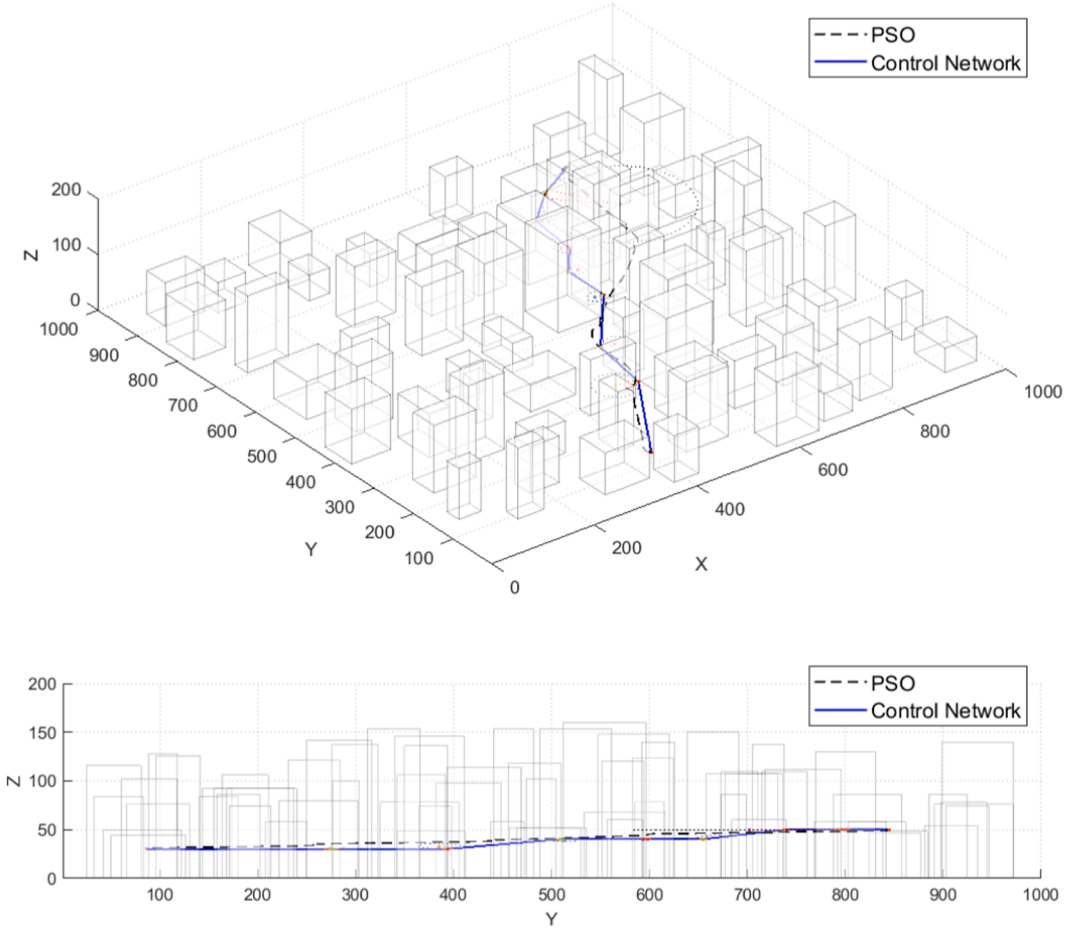


Fig. 17d. Acquired paths for OD pair 4.

the speed curve constantly shows "V" shapes during the flight process (a total of 6 decelerate-then-accelerate maneuvers), indicating that the UAV constantly experiences sharp steering. Meanwhile, the control network approach obviate such undesirable situations by the expanded control network which considers UAV kinetics during the hovering process, as discussed in Section 2.2.1.

3.3. Tests based on a real city

We choose the Yuzhong Peninsula in Chongqing as the testbed in this subsection. Two OD pairs are chosen to demonstrate our results. The 3D data of cluttered buildings and topography are extracted from OpenStreetMap (Mooney and Minghini, 2017). Latitude and longitude coordinates from online geolocation systems are translated into NED (North[x], East[y], Down[z]) cartesian coordinate system, assuming a flat earth. The extracted buildings are remodeled by using Minimum Bounding Box algorithms (Feng et al., 2020) to generate control points. The total number of buildings is 1,497, and the expanded control network contains 188,545 nodes and 17,892,562 links in this test. The overviews of the planned paths in the city of Chongqing for these two OD pairs are presented as follows:

1) OD pair 1

The planned path for OD pair 1 (from [440.31,188.53,10] to [5016.17,444.94,50]) is shown in Fig. 19a. In this case, the planned path goes over dozens of buildings and flies nearly a straight line through the gaps between cluttered buildings. The total flying time obtained by the time-optimal trajectory generation is 315.36 s, and the total computing time is 1.83 s (1.73 s for shortest path finding and 0.10 s for time-optimal trajectory generating). (SEE Fig. 19b.)

2) OD pair 2

The planned path for OD pair 2 (from [440.309,188.534,10] to [4864.713,1299.143, 70.0]) is shown in Fig. 19a. In this case, the

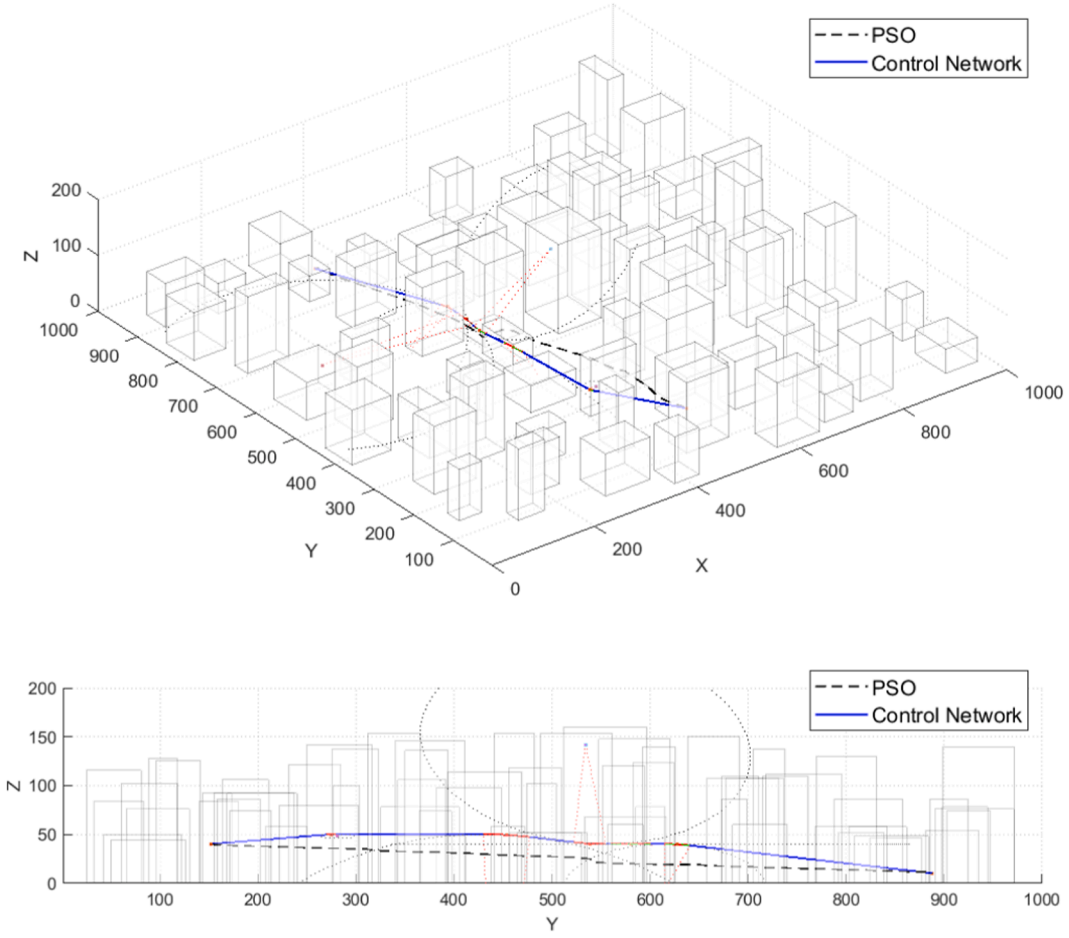


Fig. 17e. Acquired paths for OD pair 5.

Table 1
Comparisons between the control network approach and PSO approach.

OD Pair	Distance (m)	PSO			Control network approach			
		Flying Time (s)	Computing Time (s)		Flying Time (s)	Improvement	Computing Time (s)	
			Shortest path	Time-optimal trajectory			Shortest path	Time-optimal trajectory
1	1024.30	71.71	764.47	0.11	70.03	**2.40%**	0.41	0.04
2	1274.40	90.97	909.50	0.12	85.95	**5.84%**	0.52	0.05
3	962.21	66.07	765.32	0.07	65.02	**1.61%**	0.38	0.04
4	915.96	69.37	746.65	0.17	61.58	**12.66%**	0.40	0.04
5	778.05	58.83	671.92	0.10	52.84	**11.34%**	0.34	0.05

UAV flies the planned path across the area and takes a detour to obviate cluttered buildings with equal complexity comparing to OD pair 1. The total flying time obtained by the time-optimal trajectory generation is 315.57 s, and the total computing time is 2.19 s (1.86 s for shortest path finding and 0.33 s for time-optimal trajectory generating).

For OD pair 1 in Chongqing Yuzhong Peninsula area, based on the same reconstructed three-dimensional map, we tested the PSO algorithm as a benchmark for comparison. The PSO algorithm costs 31,239 s and yet failed to generate a complete path from the origin to the designated destination. Graphical illustrations of paths generated by two algorithms are shown in Fig. 20; it suggests the superior performance of the proposed control network approach in real-world-scale problem instances over PSO. Also, we observe that the total computational times (including path finding, curve smoothing and time-optimal trajectory identification) in this long-distance test are no more than 2.0 s, indicating that even for realistic-scale metropolitan areas, the computational burden of our approach is very subtle. This test demonstrates the applicability of the control network approach of UAV trajectory planning in highly complicated working environment.

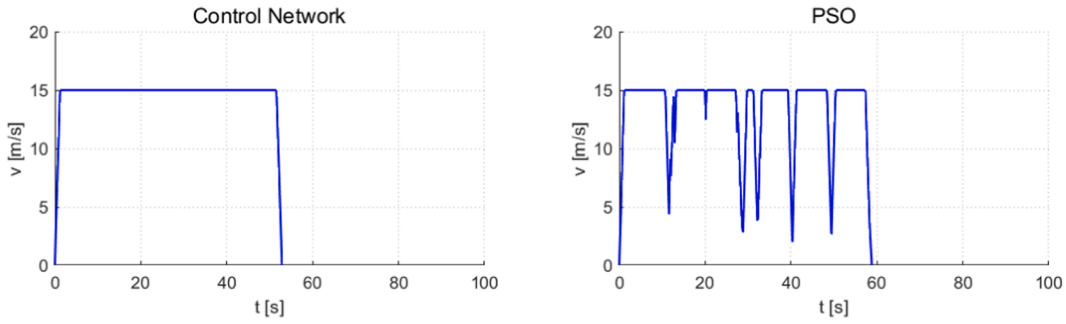


Fig. 18. Speed curves under two different approaches for OD pair 5.

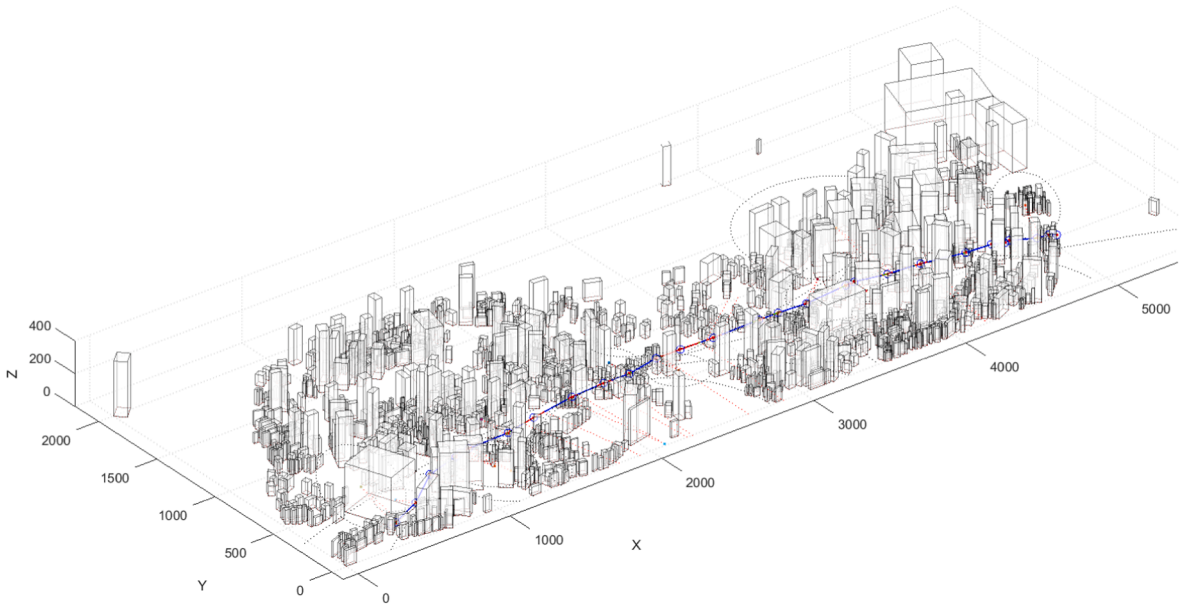


Fig. 19a. Acquired path for OD pair 1.

4. Conclusions

In this study, we introduced a control network approach for UAV trajectory planning in complicated, large-scale 3D urban environment. It contains an offline stage for control network establishment and an online stage for trajectory optimization. In the offline stage, we constructed the control network with nodes and non-conflicting links, and it was expanded to consider some kinetic properties of UAV motion. Link travel times are evaluated based on feasible UAV motion. In the online stage, the time-optimal route in the control network is identified to represent a valid "corridor", and a smoothened spatial path is acquired by Dubins curve smoothing technique. Then, a time-optimal trajectory planning algorithm is implemented for tracing the path. Numerical studies based on artificial and real-world cases demonstrated the effectiveness of the proposed methodological framework. It is demonstrated that the proposed approach was with 1.61%-12.66% flying time reduction compared to a benchmark method, i.e., PSO approach, and in a test area of Chongqing, China with 1,497 buildings the proposed method can acquire a trajectory solution within 2 s.

Some possible future research directions are suggested. First of all, the established control network (and its expansion) can be utilized in problems other than trajectory planning. For example, it can be used as the basis for "UAV salesman problem" that requires UAVs to fulfill multiple tasks in a region. The control network can also be treated as a 3D road network for UAVs in order to manage the UAV traffic, which is well suited for the future urban aerial traffic management. Secondly, establishing the control network is associated with several parameter choices, including the density of control points, the allowable link connection, and the UAV motion state differentiation on each control point; the selection is a trade-off between the complexity of the control network (which affects the computational efficiency) and the solution quality, and therefore it requires careful testing. Thirdly, for huge-scale cities, the control network could contain billions of links and it may cause the path-finding problem computationally burdensome; for alleviating this issue, appropriate partitioning of the control network is required and worthy of investigation. Lastly but not the least, how to combine

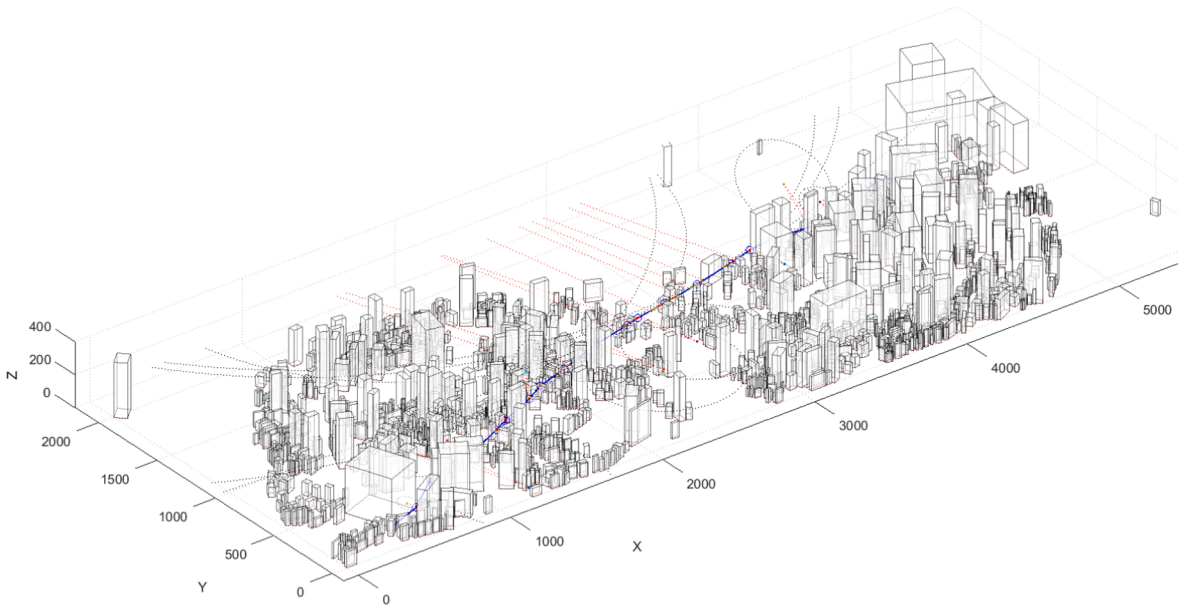


Fig. 19b. Acquired path for OD pair 2.

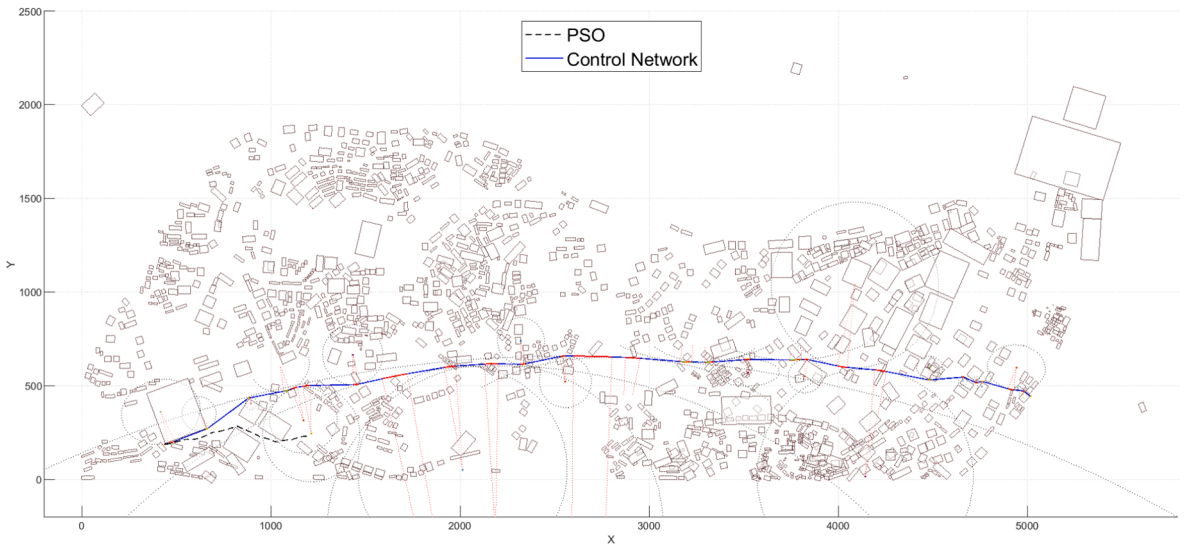


Fig. 20. Comparison of acquired trajectories with PSO method.

the control network approach with other UAV trajectory planning objectives (e.g., minimizing energy consumption), remains a challenging task to be studied in the future.

CRediT authorship contribution statement

Xi Lin: Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Writing-original draft, Writing-review & editing. **Chengzhang Wang:** Formal analysis, Methodology, Software, Writing-original draft, Writing-review & editing. **Kaiping Wang:** Methodology, Software, Writing-original draft, Writing-review & editing. **Meng Li:** Conceptualization, Funding acquisition, Supervision, Writing-review & editing. **Xiangqian Yu:** Software.

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